

Boeing 747 ("Jumbo Jet")



"The Boeing 747 is so big that it has been said that it does not fly; the Earth merely drops out from under it."

CAPTAIN NED WILSON
PAN AM PILOT

Fin built up on two-spar box structure

Rudder divided into upper and lower segments



TESTING A TURBOFAN

Test engineers place a prototype turbofan engine into a test stand at the Pratt & Whitney Aircraft Plant in Connecticut. Many of the problems during the development of the 747 concerned its engines.

Rearmost passenger exit/entry door

JUMBO CARRIER

The 747 has undergone substantial upgrading since first entering service. The 747-400 variant, shown here, was first introduced in 1989 and proved the most successful of the 747 family, with a total of 694 built before production halted in 2009. It was superseded by the 747-8, which was first flown in 2010.

A NEW DAWN

Over 210ft (64m) long and with a tail reaching six stories high, the first 747 (right) towers over several thousand guests at the roll-out ceremony in Everett on September 30, 1968. It made its maiden flight in February 1969.



aircraft, which it had lost out on after the contract went to Lockheed. The 747 was to be a passenger airliner on the same unprecedented scale. It would also be convertible to a freighter, since both Trippie and Allen suspected the future of passenger flight might lie with supersonic aviation, which would make the 747 redundant as a passenger carrier.

In Everett, north of Seattle, Boeing built an entirely new factory (larger in volume than any other building in the world) to make the giant airliner. Lack of time and capacity meant that much of the design and manufacturing had to be contracted out to other firms, with Boeing responsible for the nose and wings. It was soon obvious that the aircraft was going to be too heavy to be powered by existing Pratt & Whitney turbofan engines. Great ingenuity was put into shedding weight, including a cutback on intended passenger load, pending an engine upgrade.

The 747 prototype was due to roll out in the fall of 1968, but did not make its maiden flight until February 1969, seven weeks behind schedule. By the time it entered service in January 1970, over 150 747s had already been ordered by airlines across the world. Its teething troubles would not prevent it from being a huge commercial success for Boeing, eventually making flight available to more people than ever before.

When the wing is full of fuel, outer portions bend down and winglets are angled further outwards, increasing span by 19m (48cm)